

Function Combinations & Polynomials

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **add, subtract, multiply, and divide** functions represented by tables, graphs, or formulas.

RELATED SKILLS

I Add, subtract, multiply, and divide function outputs.

A Identify input and output values in a formula, graph, table, or context.

- ☐ 2. I can **determine** where a combination of functions is positive, zero, or undefined.

RELATED SKILLS

I Determine the domain of a sum, difference, product, quotient, or composition.

D Add, subtract, multiply, and divide function outputs.

- ☐ 3. I can **determine** polynomial end behavior from degree and leading coefficient.

RELATED SKILLS

I Infer polynomial end behavior from degree and leading coefficient.

R Identify terms, factors, coefficients, variables, and degree in a polynomial expression.

- ☐ 4. I can **identify** zeros and multiplicities and relate them to polynomial graph behavior.

RELATED SKILLS

I Relate zero multiplicity to local graph behavior.

D Determine real and complex zeros using an appropriate method.

A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.

- ☐ 5. I can **estimate** a polynomial's possible degree from its zeros, turning points, inflection points, and end behavior.

RELATED SKILLS

I Use inflection points to establish a minimum possible polynomial degree.

I Use turning points to establish a minimum possible polynomial degree.

I Use zeros and their multiplicities to establish a minimum possible polynomial degree.

D Infer polynomial end behavior from degree and leading coefficient.

D Relate zero multiplicity to local graph behavior.

☐ **6. I can **construct** a polynomial function from specified zeros, multiplicities, and a point.**

RELATED SKILLS

- ☐ **D** Build a polynomial from zeros, multiplicities, and a point.
 - ☐ **A** Determine real and complex zeros using an appropriate method.
 - ☐ **A** Relate zero multiplicity to local graph behavior.
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☐ **7. I can **match** polynomial formulas and graphs using intercepts, multiplicity, degree, and end behavior.**

RELATED SKILLS

- ☐ **D** Infer polynomial end behavior from degree and leading coefficient.
 - ☐ **A** Determine real and complex zeros using an appropriate method.
 - ☐ **A** Relate zero multiplicity to local graph behavior.
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☐ **8. I can **build and optimize** a polynomial model in a geometric or economic context.**

RELATED SKILLS

- ☐ **D** Interpret vertex and zeros in an application.
 - ☐ **A** Explain a model parameter in the language of its context.
 - ☐ **A** Restrict a model to meaningful contextual inputs.
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